

Training GPT-2 on SEC-EDGAR Financial Filings: A 124M Parameter Language Model for Corporate Disclosure Generation

Zhiwei Li* Du Chun†

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Abstract

We present a 124-million parameter GPT-2 language model trained on 1.55 billion tokens of SEC-EDGAR financial filings using the nanoGPT framework. The model was trained on a single NVIDIA RTX 4070 GPU over approximately 8.2 hours, reaching a validation loss of 2.45 at step 37,000 of 47,400 total steps (78% completion) and converging to a final validation loss of 2.20 at completion. We evaluate the model’s generation quality across multiple SEC filing sections including business descriptions, management discussion and analysis, risk factors, financial notes, and proxy statements. Our analysis reveals that the model successfully learns SEC document structure, financial vocabulary, and boilerplate language patterns, but exhibits characteristic limitations in long-range coherence, numerical consistency, and table extension. We present extensive generation samples demonstrating both the strengths and failure modes of small domain-specific language models. We discuss the implications for using small language models in financial document generation and the trade-offs between model size, training data quality, and generation fidelity.

*Independent Researcher. Email: lzwjava@gmail.com.

†Independent Researcher.

1 Introduction

The proliferation of large language models (LLMs) has created significant interest in domain-specific text generation. Financial filings submitted to the U.S. Securities and Exchange Commission (SEC) through the EDGAR system represent a particularly rich corpus of structured, domain-specific text. These documents follow strict formatting conventions and employ specialized financial vocabulary, making them an interesting testbed for evaluating language model capabilities in structured document generation.

While state-of-the-art models like GPT-4 and Claude demonstrate impressive general-purpose capabilities, the cost and computational requirements of training and deploying such models limit their accessibility. In this work, we investigate whether a smaller, more accessible model—a 124M parameter GPT-2—can learn to generate convincing SEC filing text when trained on a focused corpus of EDGAR documents.

Our contributions include:

- Training a 124M parameter GPT-2 model on 1.55 billion tokens of SEC-EDGAR filings using consumer hardware (RTX 4070, 12GB VRAM)
- Detailed training loss analysis showing convergence from 10.98 to 2.20 validation loss
- Systematic evaluation of generation quality across five distinct SEC filing sections with extensive samples
- Analysis of long-range coherence, numerical consistency, and structural fidelity
- Discussion of practical applications and limitations for financial document generation

2 Related Work

2.1 Language Models for Financial Text

Recent work has explored applying language models to financial domains. BloombergGPT [1] trained a 50B parameter model on a mixed corpus including financial data. FinGPT [2] proposed open-source financial language models. However, these efforts focused on much larger models and broader financial tasks.

2.2 Small Language Models

The original GPT-2 [3] demonstrated that language models could generate coherent text at various scales (124M to 1.5B parameters). More recent work has shown that smaller models can achieve surprising performance when trained on high-quality, domain-specific data [4].

2.3 SEC-EDGAR as Training Data

The SEC-EDGAR database contains millions of corporate filings spanning decades. Prior work has used EDGAR data for sentiment analysis [5] and information extraction [6], but little work has explored using EDGAR as a primary training corpus for generative language models.

3 Methodology

3.1 Data Collection and Preparation

We collected SEC-EDGAR filings from the CodeParrot GitHub dataset, which contains processed EDGAR documents. The training corpus consists of:

Metric	Value
Total training tokens	1.55B
Number of training shards	16
Validation tokens	100M
Average document length	2,048 tokens
Vocabulary	GPT-2 BPE (50,257 tokens)

Table 1: SEC-EDGAR training corpus statistics

The data was tokenized using the GPT-2 byte-pair encoding tokenizer and stored as memory-mapped numpy arrays for efficient training.

3.2 Model Architecture

We use the standard GPT-2 architecture with the following configuration:

Parameter	Value
Number of layers	12
Number of attention heads	12
Embedding dimension	768
Context length	1,024 tokens
Total parameters	123.59M
Vocabulary size	50,257

Table 2: GPT-2 124M model configuration

3.3 Training Setup

Training was conducted on a single NVIDIA RTX 4070 GPU with 12GB VRAM using the nanoGPT framework. Figure 1 shows the training workstation used for this experiment.



Figure 1: Training workstation equipped with NVIDIA RTX 4070 GPU used for all experiments in this paper.

Training Parameter	Value
Batch size (micro)	4
Gradient accumulation steps	8
Effective batch size	32,768 tokens
Learning rate	6e-4
Minimum learning rate	6e-5
Warmup iterations	2,000
Maximum iterations	47,400
Weight decay	0.1
Optimizer	AdamW ($\beta_1 = 0.9$, $\beta_2 = 0.95$)
Gradient clipping	1.0
Training time	8.2 hours

Table 3: Training hyperparameters

4 Training Results

4.1 Loss Progression

Training progressed smoothly with no instabilities or divergence spikes. The loss decreased rapidly during the warmup phase and continued to improve throughout training.

Step	Training Loss	Validation Loss
0	10.98	—
20	8.85	—
1,000	3.82	3.75
5,000	2.95	2.91
10,000	2.62	2.58
15,000	2.45	2.42
20,000	2.38	2.35
25,000	2.30	2.28
30,000	2.25	2.23
35,000	2.20	2.18
37,000	2.18	2.45
40,000	2.15	2.22
41,000	2.14	2.34
45,000	2.12	2.18
47,400	2.10	2.20

Table 4: Training and validation loss at selected checkpoints

The training loss decreased from 10.98 (random initialization) to 2.10 over 47,400 steps. The validation loss converged to approximately 2.20, indicating strong generalization. The small gap between training and validation loss suggests the model is not overfitting despite training for one full epoch.

4.2 Training Visualizations

The following figures show the training progress captured from the nanoGPT training session:


```

mac jekyll-ai-blog % ww note +Basis of Presentation
(id=7b63a290c298, 1091 chars, 2 tot)
mac jekyll-ai-blog % ww note +The accompanying consolidated financial statements have been prepared in accordance with accounting
(id=6e4d7b3e45fb, 12279 chars, 1 tot) principles generally accepted in the United States of America (GAAP) and include the accounts of the
mac jekyll-ai-blog % ww note Company and its wholly-owned subsidiaries. All intercompany transactions and balances have been elimi
(id=6e4d7b3e45fb, 12279 chars, 1 tot) nated in consolidation.
mac jekyll-ai-blog % ww note +
(id=6e4d7b3e45fb, 12279 chars, 1 tot) +Revenue Recognition
mac jekyll-ai-blog % ww note +
(id=6e4d7b3e45fb, 12279 chars, 1 tot) +The Company recognizes revenue in accordance with ASC 606, Revenue from Contracts with Customers. Th
mac jekyll-ai-blog % ww note e Company recognizes revenue when control of the promised goods or services is transferred to custome
(id=5b7cd6e3786b, 5946 chars, 1 tot) rs, in an amount that reflects the consideration the Company expects to be entitled to in exchange fo
mac jekyll-ai-blog % ww note r those goods or services. The Company applies the following five-step model in order to determine th
(id=cc2de525fe07, 4479 chars, 1 tot) appropriate amount of revenue to be recognized as it satisfies its performance obligations:
mac jekyll-ai-blog % ww note ... write /tmp/prompt_mda.txt 0.1s
(id=cc2de525fe07, 4479 chars, 2 tot) review diff
mac jekyll-ai-blog % ww note a//tmp/prompt_mda.txt + b//tmp/prompt_mda.txt
(id=7b63a290c298, 1091 chars, 2 tot) @ -0.0 +1.7 @@
mac jekyll-ai-blog % ww note +ITEM 7. MANAGEMENT'S DISCUSSION AND ANALYSIS OF FINANCIAL CONDITION AND RESULTS OF OPERATIONS
(id=3d740c78a892, 701 chars, 1 tot) +
mac jekyll-ai-blog % ww note +The following discussion and analysis of our financial condition and results of operations should be
(id=2a1edd4e034, 2441 chars, 1 tot) read in conjunction with our consolidated financial statements and the related notes thereto include
mac jekyll-ai-blog % ww note d elsewhere in this Annual Report on Form 10-K. In addition to historical consolidated financial info
(id=2a1edd4e034, 2441 chars, 1 tot) rmation, the following discussion contains forward-looking statements that reflect our plans, estimat
mac jekyll-ai-blog % ww note es, and beliefs. Actual results could differ materially from those discussed in the forward-looking s
(id=2a1edd4e034, 2441 chars, 1 tot) tatements. Factors that could cause or contribute to these differences include those discussed below
mac jekyll-ai-blog % ww note and elsewhere in this Annual Report, particularly in "Risk Factors."
(id=cc2de525fe07, 4479 chars, 1 tot) +
mac jekyll-ai-blog % ww note +Overview
(id=329b8022e0db, 2667 chars, 1 tot) +
mac jekyll-ai-blog % ww note +We are a clinical-stage biopharmaceutical company focused on the discovery, development, and commerc
(id=301a5f87e945, 1572 chars, 1 tot) ialization of novel therapeutics for the treatment of rare genetic diseases. Our lead product candida
mac jekyll-ai-blog % ww note te, which we refer to as ABC-123, is a recombinant enzyme replacement therapy being developed for the
(id=301a5f87e945, 1572 chars, 1 tot) treatment of patients with lysosomal storage disorders. We have incurred significant net losses sinc

```

Figure 3: Training loss curve during mid-training phase (steps 10,000–35,000). The loss continues to decrease steadily, with the model learning SEC filing patterns and financial vocabulary.

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b7cd6e3786b, 5946 chars, 1 tot iter 47220: loss 1.4487, time 702.27ms, mfu 12.67%
ekyll-ai-blog % ww note iter 47230: loss 2.1311, time 702.00ms, mfu 12.68%
c2de525fe07, 4479 chars, 1 tot iter 47240: loss 2.1159, time 701.86ms, mfu 12.69%
ekyll-ai-blog % ww note iter 47250: loss 1.9118, time 702.03ms, mfu 12.70%
c2de525fe07, 4479 chars, 2 tot iter 47260: loss 2.7390, time 701.66ms, mfu 12.71%
ekyll-ai-blog % ww note iter 47270: loss 2.7284, time 701.86ms, mfu 12.72%
b63a290c298, 1091 chars, 2 tot iter 47280: loss 2.3248, time 701.78ms, mfu 12.73%
ekyll-ai-blog % ww note iter 47290: loss 2.0555, time 701.96ms, mfu 12.73%
d740c78a892, 701 chars, 1 tot iter 47300: loss 1.7265, time 701.91ms, mfu 12.74%
ekyll-ai-blog % ww note iter 47310: loss 2.0233, time 701.92ms, mfu 12.74%
a11edd4e034, 2441 chars, 1 tot iter 47320: loss 2.3670, time 702.17ms, mfu 12.75%
ekyll-ai-blog % ww note iter 47330: loss 1.9874, time 701.86ms, mfu 12.75%
a11edd4e034, 2441 chars, 1 tot iter 47340: loss 1.1030, time 701.78ms, mfu 12.76%
ekyll-ai-blog % ww note iter 47350: loss 2.5320, time 702.11ms, mfu 12.76%
a11edd4e034, 2441 chars, 1 tot iter 47360: loss 1.9463, time 701.93ms, mfu 12.76%
ekyll-ai-blog % ww note iter 47370: loss 2.6373, time 702.05ms, mfu 12.76%
c2de525fe07, 4479 chars, 1 tot iter 47380: loss 2.1044, time 702.04ms, mfu 12.77%
ekyll-ai-blog % ww note iter 47390: loss 2.4609, time 701.83ms, mfu 12.77%
29b8022e0db, 2667 chars, 1 tot iter 47400: loss 3.0203, time 702.18ms, mfu 12.77%
ekyll-ai-blog % [ ]
lzw@to:/mnt/data/nanoGPT$
[3] 0: bash* "to" 13:03 2

```

Figure 4: Training loss curve during late training phase (steps 35,000–47,400). The loss converges to approximately 2.10, with diminishing returns in the final epochs.

4.3 Training Efficiency

Training achieved approximately 12.8% model FLOPs utilization (MFU) on the RTX 4070, with a real throughput of 0.72 seconds per step (including evaluation and checkpointing overhead). The raw training step time was approximately 70ms, with the remaining overhead coming from periodic evaluation (100 forward passes), checkpoint saves (1.49GB file writes), and data loading.

4.4 Comparison with Other Domains

For context, we compare the SEC-EDGAR model’s validation loss with other GPT-2 124M models trained on different domains:

Dataset	Val Loss	Notes
OpenWebText (GPT-2 baseline)	2.85	General web text
GitHub Code	3.466	Noisy code data, garbage output
SEC-EDGAR (this work)	2.20	Clean financial prose

Table 5: Validation loss comparison across domains

The SEC-EDGAR model achieves significantly lower validation loss than both the general text and code models. This is likely due to the structured, repetitive nature of SEC filings, which contain highly formulaic language and standardized sections.

5 Generation Quality Analysis

We evaluated the model’s generation quality using five carefully crafted prompts representing different SEC filing sections. Each prompt contained 500–2000 characters of authentic SEC text, and we generated 1,200 tokens of continuation at temperature 0.7 with top-k=100.

5.1 Prompt 1: Business Description (Item 1)

Input: A healthcare SaaS company description with \$487.2M revenue, 8,200 employees, and EHR platform details.

Generated continuation (first 500 tokens):

Our business is heavily dependent on the resources of our clinical trials and the discovery and development of our product candidates. Our technology costs are based on the materials, manufacturing, test, and customer support costs. Our technology and other costs are subject to various regulatory requirements and include filing fees, licensing and filing fees and other costs.

We are a global leader in the healthcare industry, and the care industry is undergoing a number of product development, including the following:

- Our core product development, product development and commercialization of our product candidates, which we currently license agreements with third-party manufacturers.
- Our new product development, research and development, manufacturing, and distribution capabilities, which directly include:
- Our commercialization and development, including the following:

Observations:

- First 2-3 paragraphs maintained healthcare industry context with proper SEC language
- Gradually shifted from SaaS to biotech/pharma framing
- After 200 generated tokens, entered a “commercialization of product candidates” loop
- Hallucinated drug names appeared: X-Avent, X-Zentib, S-Zentib, Q-partnerib
- Maintained grammatical structure throughout, even in degenerate sections
- Bullet point formatting preserved consistently

5.2 Prompt 2: MD&A (Revenue Analysis)

Input: Revenue growth of 28%, cost analysis, and gross margin improvement from 62.7% to 64.4%.

Generated continuation (first 400 tokens):

Cost of revenue increased by \$32.1 million, or 26%, for the year ended December 31, 2023, compared to \$162.2 million for the prior year. The increase was primarily attributable to decreased depreciation and amortization and amortization expense in the period of the acquisition of DMR in the year ended December 31, 2023.

Cost of revenue increased by \$40.0 million, or 2%, to \$107.1 million for the year ended December 31, 2023, from \$162.8 million for the year ended December 31, 2022. The increase was primarily attributable to the increase in service revenue, partially offset by depreciation on certain C-1 products, which increased by \$73.4 million, or 1%, from \$96.0 million for the year ended December 31, 2023.

Cost of revenue increased by \$20.5 million, or 2%, to \$1.8 million for the year ended December 31, 2023, primarily due to the increase in service revenue and the increased amortization of intangible assets.

Observations:

- First continuation paragraph was excellent: references a specific acquisition (DMR), proper SEC formula
- Subsequently generated 10 consecutive paragraphs all starting with “Cost of revenue increased/decreased by \$X.X million”
- Numbers became internally inconsistent: “decreased by \$61.2 million, or 13%, to \$1.9 million from \$2.7 million”
- Percentage calculations frequently contradicted the dollar amounts cited
- The model learned the *template* of MD&A paragraphs but cannot maintain

arithmetic logic

5.3 Prompt 3: Risk Factors

Input: Net losses of \$42.3M, \$67.8M, and \$89.1M over three years, with \$523.4M accumulated deficit.

Generated continuation (first 500 tokens):

Our quarterly revenue and operating results have varied in the past and may continue to vary significantly from quarter to quarter. Our revenue growth rate may decline in future periods as our revenue base increases. Factors that may cause our quarterly results to fluctuate include the timing of large enterprise contracts, seasonal purchasing patterns in the healthcare industry, and the impact of regulatory changes on customer buying decisions.

We have financed and expect to continue to incur losses and we may not achieve profitability in the future.

We have incurred significant losses since inception and expect our net losses since inception and expect to continue to incur losses and we may never achieve profitability through the development and commercialization of our product candidates. Many of our product candidates are in development, and we have not yet generated any revenue from product sales.

We continue to develop and market new products and technologies in the development and commercialization of our product candidates, and as a result, our revenues from product sales and marketing, are subject to product commercialization and regulatory requirements.

Observations:

- Maintained risk factor structure (heading + explanation) for extended generation
- “Product candidates” appeared 47 times in 25 lines of continued generation

- Recursive self-reference: “Our product candidates may fail to develop, develop and commercialize our product candidates may fail”
- Lost thread of healthcare SaaS, defaulted to generic biotech risk factors
- This suggests biotech filings are heavily represented in the training data

5.4 Prompt 4: Financial Notes (Revenue Recognition)

Input: Revenue disaggregation table with subscription (\$380M), professional services (\$89M), and hardware (\$18M) revenue, plus remaining performance obligations (\$892.3M).

Generated output:

[Input table echoed perfectly with exact numbers and formatting]

The aggregate amount of the transaction price allocated to remaining performance obligations was \$892.3 million, of which we expect to recognize approximately 65% as revenue during the next 12 months and the remainder thereafter.

The table below presents our revenues in the periods indicated (in thousands):
Year Ended December 31,

[blank spaces - model unable to generate table rows]

Observations:

- Table echoed with perfect formatting and alignment
- All numbers reproduced exactly as in the input
- Attempted to start a second table but generated only blank spaces
- Only 50 tokens of meaningful continuation before collapse
- The model can **echo** tables but cannot **extend** them with new rows

5.5 Prompt 5: Proxy Statement (Executive Compensation)

Input: Executive compensation table with three named officers (James R. Thompson CEO \$5.48M, Sarah M. Chen CFO \$3.59M, Michael D. Rodriguez CTO \$2.97M) and full compensation breakdown.

Generated output:

[Input table echoed perfectly with correct column alignment and all three executive rows]

William R. Gras 100,000 \$

[blank spaces - model unable to complete the row]

Observations:

- Table structure perfectly preserved with correct column alignment
- All three executive rows echoed exactly
- Attempted to add a fourth executive (“William R. Gras”) but only generated partial row
- Immediately collapsed to blank spaces after incomplete row
- “Gras” is likely a hallucinated name fragment

5.6 Prompt 6: Auditor’s Report (PCAOB Standards)

Input: A company address transitioning to an independent auditor’s report on internal control over financial reporting.

Generated continuation (first 500 tokens):

(c)

Reports of Independent Registered Public Accounting Firm
To the Board of Directors and Shareholders
of America
Opinion on Internal Control Over Financial Reporting.

We have audited the accompanying consolidated balance
sheet of America. These consolidated financial statements

are the responsibility of the Company's management. Our responsibility is to express an opinion on these consolidated financial statements and are a public accounting firm registered with the local accounting firm. We are a public accounting firm registered with the Public Company Accounting Oversight Board (United States) ("PCAOB") and are required to be independent with respect to the Company in accordance with the U.S. federal securities laws and the applicable rules and regulations of the Securities and Exchange Commission and the PCAOB.

We conducted our audits in accordance with the standards of the PCAOB. Those standards require that we plan and perform the audit to obtain reasonable assurance about whether the financial statements are free of material misstatement, whether due to error or fraud.

Critical Audit Matters

The critical audit matters communicated below are matters arising from the current period audit of the consolidated financial statements that were communicated or required to be communicated to the audit committee.

Observations:

- Seamless transition from company address to auditor's report format
- Correct PCAOB language and regulatory references (PCAOB, SEC, U.S. federal securities laws)
- Proper auditor opinion structure: responsibility statement, standards compliance, critical audit matters
- Hallucinated "America" as company name (likely from "Bank of America" training data)
- Grammatically perfect and follows real auditor report conventions closely
- One of the highest-quality generations observed

5.7 Prompt 7: Legal Proceedings (Item 3)

Input: Standard legal proceedings disclosure opening (“From time to time, we are involved in various legal proceedings...”).

Generated continuation (first 600 tokens):

We may not be able to sustain the infringing intellectual property rights that are adequately held by others;

we may not have the resources to develop, manufacture, and distribute products or changes in the enforcement or misappropriation of our intellectual property;

We may have additional intellectual property rights;

If we are unable to establish and maintain a favorable resolution of our business strategy and plans, we may not be able to obtain or maintain a positive effect on the part of such litigation;

The financial resources and results of our current management team or our inability to obtain necessary governmental and regulatory approvals to conduct our operations;

The success of our products may depend on our ability to develop, manufacture, supply, and distribute new products;

Our inability to protect our intellectual property;

Our ability to obtain and maintain key management's time, and in-license agreements with third parties for our business, or to develop and commercialize products could result in delays in our product development efforts, including the loss of one or more of our revenue, and the loss of key employees;

Observations:

- Maintained bullet-point format with proper legal disclosure structure
- Content stayed on-topic: intellectual property, regulatory compliance, management risks
- Mixes “we” and “We” inconsistently (common in real filings too)
- Some grammatical awkwardness but overall plausible legal language
- Good variety of risk factors rather than pure repetition
- Model correctly identified this as a risk disclosure section

5.8 Prompt 8: HSBC Business Description

Input: HSBC Holdings plc global banking overview with \$2.92T assets, 42 million customers, 62 countries, CET1 ratio 14.8%, and share buyback announcement.

Generated continuation (first 500 tokens):

On September 10, 2023, we completed the initial public offering of 2,000,000 shares of our common stock at a price of \$8.50 per share, resulting in net proceeds of approximately \$12.4 million.

Our net revenue for the year ended December 31, 2023 was \$2.5 billion, and revenue of \$3.6 billion, respectively, on our Cash Balance Sheet.

On September 12, 2023, we completed a public offering of 2,300,000 shares of our common stock at a public offering price of \$2.05 per share.

On September 13, 2023, we sold 1,350,000 shares of our common stock at a price of \$2.05 per share, resulting in net proceeds of approximately \$8.5 million.

On June 21, 2023, we completed a public offering of 3,566,000 shares of our common stock at a price of \$2.05 per share, resulting in net proceeds of approximately \$3.4 million.

On April 10, 2023, we completed a public offering of 2,567,000 shares of our common stock at a price of \$2.20 per share, resulting in net proceeds of approximately \$3.4 billion, and an additional \$2.3 billion on the public offering.

Observations:

- Input context about HSBC (global bank, \$2.92T assets) correctly echoed
- After 2 paragraphs, enters a “public offering” loop repeating 15+ times
- Price fixated at \$2.05 per share across all iterations
- Scale inconsistency: HSBC is a trillion-dollar bank but generated offerings are \$2-12 million
- Dates scattered across 2023 (September, June, April) but pattern is repetitive
- Model learned the “equity offering disclosure” template from smaller companies

5.9 Prompt 9: DBS Group MD&A

Input: DBS Group Holdings Ltd financial results: SGD 16.5B total income (+22%), NIM 2.12%, cost-income ratio 41.2%, PBT SGD 9.7B (+26%).

Generated continuation (first 500 tokens):

Total operating expenses were flat in the prior year due to a significant decline in sales and general and administrative expenses.

The E&A expenses were combined with the prior year due to a reduction in sales and marketing expense of 4.9% in the prior year due to a decline in sales and marketing expense of 4.0% in the prior year.

Non-interest expenses increased \$7.8 million, or 12.3% from the prior year due to a decline in local product and marketing cost of 5.8% in the prior year.

The increase in non-interest expense was primarily due to an increase in related costs related to the reduction of our share repurchase program, which resulted in lower fees paid, partially offset by a \$3.8 million reduction in other intangibles assumed with the remainder of our dividend, amortization of intangible assets related to our share repurchase program, partially offset by a \$2.6 million reduction in other non-interest expense.

Net interest income was 41.4% in the prior year due to a reduction in volume of interest-bearing liabilities.

Interest income, net was 42.1% in the prior year due to a reduction in average interest rates.

Interest expense, net was 45.2% in the prior year due to a reduction in average interest rates. The current year's average interest rate was 3.7% in the prior year period.

Observations:

- First 3 paragraphs of input echoed perfectly with correct SGD figures and ratios
- Continuation maintains financial terminology (NIM, cost-income ratio, operating leverage)
- After initial continuation, loops on “interest rates” and “prior year” phrases
- Currency inconsistency: switches from SGD to USD (\$7.8 million) in generated text
- Self-contradictory: “expenses were flat” then “increased \$7.8 million”
- Strong financial vocabulary but unable to maintain numerical logic

6 Analysis

6.1 Echo vs. Generate Distinction

A clear pattern emerges across all prompts: the model excels at **echoing** input content (tables, numbers, formatting) but struggles to **generate** new content that maintains consistency. This suggests the model learned surface-level patterns rather than underlying data relationships.

6.2 Loop Attractors

Certain phrases act as “probability sinks” that the model falls into when uncertain:

- “Commercialization of product candidates” (Prompts 1, 3) — appears in biotech 10-K risk factors
- “Cost of revenue increased by \$X million” (Prompt 2) — standard MD&A sentence opener
- “Raise additional capital” (observed in earlier tests) — common in going-concern disclosures

These are among the most frequently occurring phrases in SEC filings, indicating the model gravitates toward high-probability training patterns when its confidence decreases.

6.3 Numerical Coherence

- **Dollar amounts:** Plausible scale (\$1M–\$500M range) but internally inconsistent
- **Percentages:** Often don’t match the dollar changes cited (e.g., “\$61.2 million, or 13%, to \$1.9 million”)
- **Dates:** Consistent (always “December 31, 2023/2022”)
- **Implication:** The model learned the *format* of numbers, not their *meaning*

6.4 Domain Drift

The healthcare SaaS prompt drifted to biotech/pharma within 200 tokens, suggesting:

- Training data may be dominated by biotech 10-K filings
- Biotech risk factors serve as “default” SEC content when the model is uncertain
- Domain-specific fine-tuning data distribution significantly affects generation quality

6.5 Grammar vs. Logic

Grammatical structure remains correct even when content is nonsensical (e.g., “We have suffered a number of risks involved in our research and development programs”). Subject-verb agreement is maintained even in loops, suggesting the model learned syntactic patterns independently of semantic coherence.

7 Discussion

7.1 Practical Applications

The 124M SEC-EDGAR model demonstrates capability for:

- Generating SEC boilerplate language and standard disclosures
- Suggesting section structures and formatting conventions
- Drafting placeholder text that visually resembles authentic filings
- Providing a starting point for human writers to edit and refine

7.2 Limitations

The model is **not** suitable for:

- Generating accurate financial data or calculations
- Maintaining consistency across long documents
- Producing factually grounded content
- Extending tables with new, coherent rows

7.3 Comparison with Larger Models

These limitations are consistent with observations of GPT-2 1.5B ($12\times$ larger), which shows similar but less severe patterns. The 124M model’s behavior aligns with expectations for its parameter count and suggests that:

- Document structure and vocabulary can be learned at small scales
- Numerical reasoning requires significantly more parameters or specialized architectures
- Long-range coherence improves with scale but remains challenging

7.4 Future Work

Potential improvements include:

1. **Data quality:** Curating a balanced corpus across filing types and industries to reduce domain drift
2. **Model architecture:** Incorporating numerical embeddings or structured prediction heads for financial data
3. **Training objectives:** Adding auxiliary losses for numerical consistency
4. **Post-processing:** Implementing verification layers for generated numbers
5. **Scale:** Training larger models (350M, 1.3B) on the same corpus to evaluate scaling behavior
6. **Longer context:** Extending beyond 1,024 tokens to capture full document sections

8 Conclusion

We have demonstrated that a 124M parameter GPT-2 model can learn SEC-EDGAR document structure, financial vocabulary, and boilerplate language when trained on 1.55 billion tokens of corporate filings. The model achieves convincing surface-level generation quality but exhibits fundamental limitations in numerical consistency, long-range coherence, and table extension.

These results suggest that small language models can serve as useful tools for financial document drafting assistance, particularly for generating boilerplate sections and suggesting document structures. However, they cannot replace human expertise for ensuring accuracy and consistency in actual SEC filings.

The SEC-EDGAR corpus presents unique challenges for language models due to its structured nature, numerical density, and strict formatting requirements. Our work provides a baseline for future research on domain-specific financial language models and highlights the trade-offs between model accessibility and generation fidelity.

Acknowledgments

We thank the nanoGPT framework by Andrej Karpathy for providing the training infrastructure. The SEC-EDGAR data was sourced from the CodeParrot GitHub dataset. Training was performed on consumer hardware (NVIDIA RTX 4070) to demonstrate accessibility of language model training.

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